

Def. Given a space X , a covering space of X consists of a space $\tilde{X}$ and a map $p: \tilde{X} \rightarrow X$ s.t.
 For each $x \in X$, there is an open neighborhood U of x in X satisfying

$$p^{-1}[U] = \bigsqcup_{i \in \mathbb{N}} \mathcal{O}_i \quad \text{and} \quad p|_{\mathcal{O}_i}: \mathcal{O}_i \xrightarrow{\cong} U_i. \quad (\text{Such } U \text{ is called "evenly covered"})$$

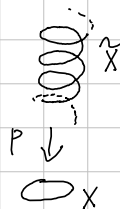
Example. Given a space S^1 , let $p: \mathbb{R} \rightarrow S^1$ s.t. $p(t) = (\cos 2\pi t, \sin 2\pi t)$.

Let $(1,0) \in S^1$ be given. Then, for a neighborhood $B_1(1,0) \cap S^1$,

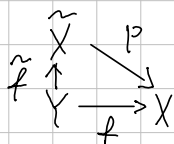
$$p^{-1}[B_1(1,0) \cap S^1] = \bigsqcup_{n \in \mathbb{Z}} (n + (-\frac{1}{8}, \frac{1}{8}))$$

and for each $n \in \mathbb{Z}$,

$$(n + (-\frac{1}{8}, \frac{1}{8})) \cong B_1(1,0) \cap S^1 \text{ via } p.$$



Def. Given a covering $p: \tilde{X} \rightarrow X$ and a map $f: Y \rightarrow X$,
 a continuous map $\tilde{f}: Y \rightarrow \tilde{X}$ such that $p \circ \tilde{f} = f$ is called a lift of f to $\tilde{X}$,
 with respect to p .



Thm. Given a map $F: Y \times I \rightarrow X$ and a map $\tilde{F}: Y \times \{0\} \rightarrow \tilde{X}$ lifting $F|_{Y \times \{0\}}$, there is a unique map $\tilde{F}: Y \times I \rightarrow \tilde{X}$ lifting F extended $\tilde{F}: Y \times \{0\} \rightarrow \tilde{X}$.
Proof.

Def. Let $p: E \rightarrow B$ be a covering map.

Given a continuous map $f: X \rightarrow B$, a lifting $\tilde{f}: X \rightarrow E$ of f is a map satisfying $p \circ \tilde{f} = f$.

Thm. Let $p: E \rightarrow B$ be a covering and $p(e_0) = b_0$.

For any path $f: [0, 1] \rightarrow B$ with $f(0) = b_0$,

there exists a unique path $\tilde{f}: [0, 1] \rightarrow E$ satisfying $p \circ \tilde{f} = f$

Proof, ~~불가능~~ 증명.

For any associated map by Homotopy of paths $F: [0, 1] \times [0, 1] \rightarrow B$ with $F(0, 0) = b_0$,

there exists a unique associated map by Homotopy of paths $\tilde{F}: [0, 1] \times [0, 1] \rightarrow E$

satisfying $p \circ \tilde{F} = F$ and $\tilde{F}(0, 0) = e_0$.